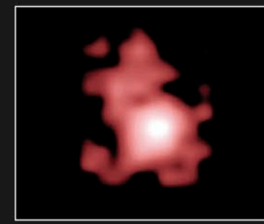
EARLY GALAXIES

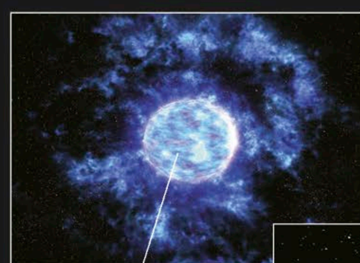
Astronomers are still trying to pinpoint when the very first stars ignited and in what types of early galactic structures this may have occurred. Recent infrared studies, with instruments such as the Spitzer Space Telescope and Very Large Telescope, have revealed what seem to be very faint galaxies with extremely high red shifts existing as little as 400 million years after the Big Bang. Their existence indicates that well-developed precursor knots and clumps of condensing matter may have existed as little as 100 to 300 million years after the Big Bang. It is within these structures that the first stars probably formed.

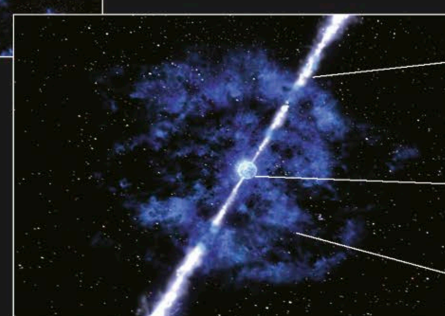


MOST DISTANT GALAXY
This image from the Hubble telescope shows GN-Z11, the most distant galaxy known, appearing as it existed just 400 million years after the Big Bang.

THE FIRST STARS

The first stars, which may have formed only 180 million years after the Big Bang, were made almost entirely of hydrogen and helium—virtually no other elements were present. Physicists think that star-forming nebulae that lacked heavy elements condensed into larger gas clumps than those of today. Stars forming from these clumps would have been very large and hot, with perhaps 100 to 1,000 times the mass of the Sun. Many would have lasted only a few million years before dying as supernovae. Ultraviolet light from these stars may have triggered a key moment in the universe's evolution—the reionization of its hydrogen, turning it from a neutral gas back into the ionized (electrically charged) form seen today. Alternatively, radiation from quasars (see p.320) may have reionized the universe.





DEATH OF MEGASTARS
The first, massive stars may have exploded as "hypernovae"—events associated today with black hole formation and violent bursts of gamma rays. These artist's impressions depict one model of hypernova development.

IONIZING POWER OF STARS
These young, high-mass stars in the Orion Nebula ionize the gas around them, causing it to glow. Ionized hydrogen between galaxy clusters today may have been created by the far fiercer radiation of the first generation of stars and hypernovae.

COSMIC CHEMICAL ENRICHMENT

During the course of their lives and deaths, the first massive stars created and dispersed new chemical elements into space and into other collapsing protogalactic clumps. A zoo of new elements, such as carbon, oxygen, silicon, and iron, was formed from nuclear fusion in the hot cores of these stars. Elements heavier than iron, such as barium and lead, were formed during their violent deaths. Second- and third-generation stars, smaller than the primordial megastars, later formed from the enriching interstellar medium. These stars created more of the heavier elements and returned them to the interstellar medium via stellar winds and supernova explosions. Galactic mergers and the stripping of gas from galaxies (see p.327) led to further intergalactic mixing and dispersion. These processes of recycling and enrichment of the cosmos continue today. In the Milky Way Galaxy, the new heavier elements have been essential to the formation of objects from rocky planets to living organisms.

STARDUST
Supernova remnant SN 1006 is a sphere of enriched material expanding into space. Elements heavier than iron have mostly been made and dispersed by supernovae.

ABUNDANCE OF ELEMENTS
The ordinary (atom-based) matter of the universe initially consisted of hydrogen and helium, with a trace of lithium. Today, it still consists mainly of hydrogen and helium, but stellar processes have boosted the contribution from other elements.

INTRODUCTION